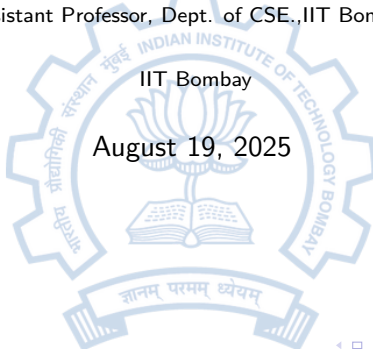


Convergence of Stochastic Gradient Descent

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1. Convergence of SGD
2. Summary of SGD Convergence
3. SGD rate improvement
4. Example: Randomised Kaczmarz
5. Variance Reduction



Theorem (Convergence of Stochastic (Sub)Gradient Descent)



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Then, for any sequence of positive step sizes $\{\alpha_k\}_{k=1}^T$, the following bound holds:

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Update Rule:

$$x_{k+1} = x_k - \alpha_k G_k = x_k - \alpha_k (\nabla f(x_k) + w_k)$$



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Applying the identity:

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- Under these two conditions, **SGM converges**.
- Convergence bound:

$$\mathbb{E}[f(\bar{x}_T) - f(x^*)] \leq O\left(\frac{BR}{\sqrt{T}}\right)$$

where $\bar{x}_T = \frac{\sum_{k=1}^T \alpha_k x_k}{\sum_{k=1}^T \alpha_k}$ and R denotes initialization distance $\|x_0 - x^*\|$.



Summary of SGD Convergence

We

- $G(x; \xi)$: Stochastic Gradient, x : Query Point, ξ : Random Variable (Noise)
- **Unbiasedness**: $\Rightarrow \mathbb{E}_{\xi}[G(x; \xi)] = \nabla f(x)$
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PL-Condition and SGD rate Improvement





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System of n -linear equations

$$Ax = b \quad \text{where } A = \begin{bmatrix} a_1^\top \\ a_2^\top \\ \vdots \\ a_n^\top \end{bmatrix}_{n \times d \text{ matrix}} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}_{n \times 1}$$



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Remark: This satisfies (AB) condition with $A^2 = M^2$, $B^2 = 0$.



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Remarks

Remark: Optimizing contracting factor over α^k , we get $\alpha^k = \frac{\mu}{A^2}$

With this α^k , we have

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 - **Answer:** $\Rightarrow$ Yes
- **Corollary:** $(0, B)$ -Bound on μ -PL Loss

$$\mathbb{E}_{\xi} \|G(x; \xi)\|^2 \leq B^2$$



Using main Theorem.. With $(0, B)$ bound and μ -PL condition, We get

$$\Delta^{k+1} \leq (1 - 2\alpha^k \mu) \Delta^k + (\alpha^k)^2 B^2$$



Lemma

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Rate: $O\left(\frac{\log(T)}{T}\right)$ instead of $O\left(\frac{1}{\sqrt{T}}\right)$ as before.

We use the PL-condition crucially.



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Usually, r is treated as a **hyper-parameter**.



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SVRG is epoch based algorithm.





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Hence SVRG is unbiased.



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Remark: This lemma implies (A,B) bound with $B=0$.



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Choose, $N = 64 \frac{M}{m}$

Thank You!